

Riya Mehta

Student ID: 2026078

Data Structures and Algorithms – Midterm Examination

Question 1 (5 Marks):

What is the time complexity of Binary Search? Explain briefly.

Answer:

The time complexity of Binary Search is $O(\log n)$.

Binary Search works only on a sorted array. In every iteration, the algorithm compares the target value with the element at the middle index.

If the target matches the middle element, the search ends immediately.

If the target is smaller, the algorithm continues searching in the left half of the array. If the target is larger, it continues in the right half.

Because one half of the remaining elements is discarded after each comparison, the search interval becomes smaller very quickly. The number of iterations required therefore increases logarithmically with the size of the array.

Hence, the overall time complexity of Binary Search is $O(\log n)$.